

BIOLOGICAL CLASSIFICATION

What is Biological Classification?

Biological classification is the scientific process of arranging living organisms into groups based on their shared characteristics and evolutionary relationships. This process has undergone significant changes over time, evolving from simple observations to more complex, scientifically-based systems. Aristotle was among the earliest to attempt a more scientific basis for classification. He categorized plants into trees, shrubs, and herbs based on simple morphological characters. For animals, he divided them into two groups: those with red blood and those without.

• The Two Kingdom System

During Linnaeus' time, a Two Kingdom system of classification was developed, which included all organisms under either Plantae or Animalia.

- Limitations of the Two Kingdom System:
 - It did not distinguish between eukaryotes and prokaryotes.
 - It did not differentiate between unicellular and multicellular organisms.
 - It failed to separate photosynthetic organisms (like green algae) from non-photosynthetic organisms (like fungi).
 - While easy to understand, a large number of organisms did not fit neatly into either category, making the system inadequate in the long run. For example, earlier systems grouped bacteria, blue-green algae, fungi, mosses, ferns, gymnosperms, and angiosperms all under 'Plants' because they all had cell walls, despite vast differences in other characteristics. This grouped prokaryotic bacteria with eukaryotic organisms, and unicellular organisms with multicellular ones (e.g., Chlamydomonas and Spirogyra were both placed under algae). It also didn't distinguish between heterotrophic fungi and autotrophic green plants, even though

their cell walls differed in composition (fungi had chitin, green plants had cellulose).

- Evolution Towards More Comprehensive Systems

- The inadequacy of older systems led to a need for incorporating other characteristics for classification, such as cell structure, nature of the cell wall, mode of nutrition, habitat, methods of reproduction, and evolutionary relationships.
- While the plant and animal kingdoms have remained constant categories across different systems, the understanding of which groups belong within them has changed. Similarly, the number and nature of other kingdoms have been interpreted differently by scientists over time.
- The overarching goal has been to develop a classification system that reflects not only morphological, physiological, and reproductive similarities, but also phylogenetic relationships, meaning it is based on evolutionary history.

- R.H. Whittaker's Five Kingdom Classification (1969)

- R.H. Whittaker proposed a Five Kingdom Classification system in 1969.
- The five kingdoms defined by him are: Monera, Protista, Fungi, Plantae, and Animalia.
- Main Criteria for Classification: Whittaker's system used several key criteria:
 - Cell structure
 - Body organisation
 - Mode of nutrition
 - Reproduction
 - Phylogenetic relationships

- Addressing Previous Issues: This system resolved many

problems of earlier classifications:

- Fungi were placed in a separate Kingdom Fungi due to their chitin cell walls and heterotrophic (saprophytic/parasitic) mode of nutrition.
- All prokaryotic organisms were grouped under Kingdom Monera.
- All single-celled eukaryotic organisms were placed in Kingdom Protista. This brought together organisms like Chlamydomonas and Chlorella (previously in Algae within Plants) with Paramecium and Amoeba (previously in the animal kingdom), demonstrating how criteria changes lead to reclassification.

Kingdom Monera: -

Sole Members: Bacteria are the sole members of the Kingdom Monera.

- **Abundance and Habitat:** They are the most abundant micro-organisms and are found almost everywhere, with hundreds present in a handful of soil. Bacteria can live in extreme habitats such as hot springs, deserts, snow, and deep oceans where few other life forms can survive. Many also exist as parasites in or on other organisms.
- **Cell Type:** Organisms in Kingdom Monera are prokaryotic.
- **Cell Wall:** They possess a non-cellulosic cell wall, composed of polysaccharides and amino acids.
- **Nuclear Membrane:** A nuclear membrane is absent in Monerans.
- **Body Organization:** Their body organization is cellular.
- **Metabolic Diversity:** Though their structure is simple, bacteria are very complex in behaviour and show the most extensive metabolic diversity compared to many other organisms.
- **Mode of Nutrition:** They can be autotrophic (synthesizing

their own food from inorganic substrates, either chemosynthetic or photosynthetic) or heterotrophic (depending on other organisms or dead organic matter, being saprophytic or parasitic).

- **Reproduction:** Bacteria mainly reproduce by fission. Under unfavourable conditions, they may produce spores. They also exhibit a primitive type of sexual reproduction involving DNA transfer from one bacterium to another.

Classification within Monera:

Bacteria are further grouped based on their shape:

- Spherical: Coccus (pl.: cocci)
- Rod-shaped: Bacillus (pl.: bacilli)
- Comma-shaped: Vibrium (pl.: vibrio)
- Spiral: Spirillum (pl.: spirilla)

The kingdom includes two major sub-groups, Archaeobacteria and Eubacteria, as well as Mycoplasma:

1. Archaeobacteria:

- These are special bacteria that live in some of the harshest habitats.
- Examples of such habitats include extreme salty areas (halophiles), hot springs (thermoacidophiles), and marshy areas (methanogens).
- Their ability to survive in extreme conditions is due to a different cell wall structure compared to other bacteria.
- Methanogens are found in the gut of ruminant animals like cows and buffaloes and are responsible for producing methane (biogas) from their dung.

2. Eubacteria ('True Bacteria'):

- Characterized by the presence of a rigid cell wall.
- If motile, they possess a flagellum.

- **Cyanobacteria (blue-green algae)** are a type of Eubacteria that are photosynthetic autotrophs, containing chlorophyll a similar to green plants. They can be unicellular, colonial, or filamentous, found in freshwater, marine, or terrestrial environments. Their colonies are often surrounded by a gelatinous sheath and can form blooms in polluted water bodies. Some cyanobacteria, like Nostoc and Anabaena, can fix atmospheric nitrogen in specialized cells called heterocyst.

- **Chemosynthetic autotrophic bacteria** oxidize various inorganic substances (such as nitrates, nitrites, and ammonia) and use the released energy for ATP production. They play a great role in recycling nutrients like nitrogen, phosphorus, iron, and sulfur.

- **Heterotrophic bacteria** are the most abundant in nature and are mostly important decomposers.

- They have significant impact on human affairs, being helpful in making curd from milk, producing antibiotics, and fixing nitrogen in legume roots.

- However, some are pathogens causing diseases in humans, crops, and animals, such as cholera, typhoid, tetanus, and citrus canker.

3. Mycoplasma:

- These organisms completely lack a cell wall.

- They are the smallest living cells known.

- They can survive without oxygen.

- Many Mycoplasmas are pathogenic in animals and plants.

This comprehensive overview from the sources highlights the diverse nature and critical roles of organisms within the Kingdom Monera.

Kingdom Protista

Kingdom Protista is one of the five kingdoms proposed by R.H. Whittaker in his Five Kingdom Classification. This kingdom was established to group together all single-celled eukaryotes. While the boundaries of this kingdom are noted as "not well defined," it includes diverse groups such as Chrysophytes, Dinoflagellates, Euglenoids, Slime moulds, and Protozoans.

Key Characteristics of Protista:

- **Cell Type:** Protists are eukaryotic, meaning their cells contain a well-defined nucleus and other membrane-bound organelles.
- **Cell Wall:** A cell wall is present in some protists, but absent in others. For example, *Chlamydomonas* and *Chlorella* (earlier placed in Algae) have cell walls, while *Paramecium* and *Amoeba* (earlier placed in the animal kingdom) lack cell walls. Euglenoids, specifically, have a protein-rich layer called pellicle instead of a cell wall, which gives their body flexibility. Diatoms have cell walls embedded with silica that form two thin overlapping shells, making them indestructible. Dinoflagellates have stiff cellulose plates on their outer surface.
- **Nuclear Membrane:** A nuclear membrane is present.
- **Body Organization:** Their body organization is cellular.
- **Mode of Nutrition:** Protists exhibit diverse modes of nutrition. They can be autotrophic (photosynthetic) or heterotrophic. For instance, Chrysophytes and Dinoflagellates are mostly photosynthetic, while Euglenoids are photosynthetic in sunlight but behave as heterotrophs by predating on smaller organisms when deprived of light. Slime moulds are saprophytic protists, and all protozoans are heterotrophs, living as predators or parasites.
- **Locomotion:** Some protists possess flagella or cilia for movement. Many Dinoflagellates have two flagella, one longitudinal and one transverse. Euglenoids also have two

flagella. Ciliated protozoans, like *Paramecium*, move actively due to thousands of cilia.

- **Reproduction:** Protists reproduce both asexually and sexually, involving a process of cell fusion and zygote formation.

- **Habitat:** Members of Protista are primarily aquatic. They are found in fresh water, marine environments, and moist soil.

- **Evolutionary Link:** This kingdom forms a link with other kingdoms, including plants, animals, and fungi.

Major Groups within Protista:

- **Chrysophytes:** This group includes diatoms and golden algae (desmids). They are found in freshwater and marine environments, are microscopic, and float passively as plankton. Most are photosynthetic. Diatoms have cell walls made of two overlapping silica shells, which are indestructible and accumulate over time to form 'diatomaceous earth', used in polishing and filtration. Diatoms are considered the chief 'producers' in the oceans.

- **Dinoflagellates:** Mostly marine and photosynthetic, they can appear yellow, green, brown, blue, or red due to their pigments. They have a cell wall with stiff cellulose plates and usually two flagella. Some, like *Gonyaulax*, can multiply rapidly, causing "red tides" and releasing toxins that may kill marine animals.

- **Euglenoids:** Majority are freshwater organisms found in stagnant water. They have a protein-rich pellicle instead of a cell wall, making their body flexible. They possess two flagella. While photosynthetic in sunlight, they become heterotrophic predators in its absence. Their pigments are similar to those in higher plants.

- **Slime Moulds:** These are saprophytic protists that move along decaying twigs and leaves, engulfing organic material. Under suitable conditions, they form an aggregation called

plasmodium, which can grow large. Under unfavourable conditions, the plasmodium differentiates into fruiting bodies that bear spores with true walls, which are highly resistant and dispersed by air currents.

- **Protozoans:** All protozoans are heterotrophs and live as predators or parasites. They are considered primitive relatives of animals.

- **Amoeboid protozoans:** Live in fresh water, sea water, or moist soil. They move and capture prey using pseudopodia (false feet). Marine forms have silica shells. Entamoeba is an example of a parasitic form.

- **Flagellated protozoans:** Can be free-living or parasitic, possessing flagella. Parasitic forms like Trypanosoma cause diseases such as sleeping sickness.

- **Ciliated protozoans:** Aquatic organisms that move actively with thousands of cilia. They have a gullet to steer food into the cell. Paramecium is an example.

- **Sporozoans:** This group includes diverse organisms with an infectious spore-like stage in their life cycle. The most well-known is Plasmodium (malarial parasite), which causes malaria.

Kingdom Fungi

Fungi constitute a unique kingdom of heterotrophic organisms and exhibit great diversity in morphology and habitat. They are cosmopolitan, found in air, water, soil, and on animals and plants, preferring warm and humid conditions.

Key Characteristics of Fungi:

- **Cell Type:** Fungi are eukaryotic.

- **Cell Wall:** Their cell walls are present and composed of chitin and polysaccharides. This distinguishes them from green plants

which have cellulosic cell walls.

- **Nuclear Membrane:** A nuclear membrane is present.

- **Body Organization:** Most fungi, with the exception of unicellular yeasts, are filamentous. Their bodies consist of long, slender, thread-like structures called hyphae, which form a network known as mycelium. Some hyphae are continuous tubes with multinucleated cytoplasm (coenocytic hyphae), while others have septate or cross walls. Body organization can be multicellular or loose tissue.

- **Mode of Nutrition:** Most fungi are heterotrophic.

- **Saprophytes:** They absorb soluble organic matter from dead substrates.

- **Parasites:** They depend on living plants and animals.

Examples include the parasitic fungus causing white spots on mustard leaves or *Puccinia* (wheat rust).

- **Symbionts:** They can live in mutually beneficial associations.

- **Lichens:** Association with algae (phycobiont), where algae prepare food and fungi provide shelter and absorb nutrients.

- **Mycorrhiza:** Association with roots of higher plants.

- **Reproduction:** Fungi reproduce both asexually and sexually, as well as by vegetative means.

- **Vegetative Reproduction:** Includes fragmentation, fission, and budding.

- **Asexual Reproduction:** By spores called conidia, sporangiospores, or zoospores.

- **Sexual Reproduction:** Involves oospores, ascospores, and basidiospores, produced in distinct fruiting bodies. The sexual cycle typically involves three steps:

1. **Plasmogamy:** Fusion of protoplasm between two motile or non-motile gametes.

2. **Karyogamy:** Fusion of two nuclei.

3. **Meiosis:** Occurs in the zygote, resulting in haploid spores.

- In some fungi, plasmogamy immediately leads to diploid cells. In others (Ascomycetes and Basidiomycetes), an intervening dikaryotic stage ($n+n$, two nuclei per cell), called a dikaryon, occurs before nuclear fusion.

Classes of Kingdom Fungi:

• Phycomycetes:

- **Habitat:** Aquatic habitats, decaying wood in moist places, or as obligate parasites on plants.
- **Mycelium:** Aseptate and coenocytic.
- **Asexual Reproduction:** By motile zoospores or non-motile aplanospores, produced endogenously in sporangium.
- **Sexual Reproduction:** A zygospore is formed by the fusion of two gametes, which can be similar (isogamous) or dissimilar (anisogamous or oogamous).
- **Examples:** Mucor, Rhizopus (bread mould), and Albugo (parasitic fungus on mustard).

• Ascomycetes (Sac-fungi):

- **General:** Mostly multicellular (e.g., Penicillium), but rarely unicellular (e.g., yeast - Saccharomyces).
- **Nutrition:** Saprophytic, decomposers, parasitic, or coprophilous (growing on dung).
- **Mycelium:** Branched and septate.
- **Asexual Reproduction:** By conidia, produced exogenously on conidiophores. Conidia germinate to produce mycelium.
- **Sexual Reproduction:** By ascospores, produced endogenously in sac-like asci. These asci are arranged in fruiting bodies called ascocarps.
- **Examples:** Aspergillus, Claviceps, Neurospora (used in biochemical and genetic work), edible morels, and truffles.

• Basidiomycetes:

- **Common Forms:** Mushrooms, bracket fungi, or puffballs.
- **Habitat:** Grow in soil, on logs, tree stumps, and as parasites

in living plants (e.g., rusts and smuts).

- **Mycelium:** Branched and septate.
- **Asexual Reproduction:** Generally not found, but vegetative reproduction by fragmentation is common.
- **Sexual Reproduction:** Sex organs are absent. Plasmogamy occurs through the fusion of two vegetative or somatic cells of different strains, leading to a dikaryotic structure that ultimately forms a basidium. Karyogamy and meiosis take place in the basidium, producing four basidiospores exogenously. Basidia are arranged in fruiting bodies called basidiocarps.
- **Examples:** Agaricus (mushroom), Ustilago (smut), and Puccinia (rust fungus).

• **Deuteromycetes (Imperfect Fungi):**

- **Distinguishing Feature:** Commonly known as imperfect fungi because only their asexual or vegetative phases are known. If their sexual forms are discovered, they are moved to the appropriate classes (often Ascomycetes or Basidiomycetes).
- **Reproduction:** Reproduce only by asexual spores known as conidia.
- **Mycelium:** Septate and branched.
- **Nutrition:** Some are saprophytes or parasites, but a large number are decomposers of litter and aid in mineral cycling.
- **Examples:** Alternaria, Colletotrichum, and Trichoderma

KINGDOM ANIMALIA

Cell Type: Organisms in Kingdom Animalia are eukaryotic. This means their cells have a well-defined nucleus and other membrane-bound organelles.

• **Cell Wall:** A defining feature is that their cells lack cell walls. This distinguishes them from plants (which have cellulose cell walls) and fungi (which have chitin and polysaccharide cell walls).

- **Nuclear Membrane:** A nuclear membrane is present in their cells.
- **Body Organization:** Animals are multicellular organisms. Their body organization can range from tissue to organ to organ system levels.
- **Mode of Nutrition:** They are heterotrophic. They depend directly or indirectly on plants for food. Their primary mode of nutrition is holozoic, which involves the ingestion of food. They digest their food in an internal cavity. Some listed examples of heterotrophic modes include holozoic and saprophytic, though the primary description for animals emphasizes ingestion.
- **Food Storage:** They store food reserves as glycogen or fat.
- **Growth and Development:** Animals follow a definite growth pattern and grow into adults that have a definite shape and size. Sexual reproduction involves the copulation of male and female, followed by embryological development.
- **Locomotion and Sensory Systems:** Higher forms of animals exhibit elaborate sensory and neuromotor mechanisms. Most animals are capable of locomotion.

Kingdom Plantae

Kingdom Plantae is one of the five kingdoms in R.H. Whittaker's classification system. It has been a constant kingdom under all different classification systems, although the understanding of which groups or organisms should be included has changed over time.

Key characteristics and details of Kingdom Plantae include:

- **Cell Type:** Organisms in Kingdom Plantae are eukaryotic. Their cells have a well-defined nucleus and other membrane-bound organelles.
- **Chlorophyll:** They are generally chlorophyll-containing organisms, which means they are primarily autotrophic.

- **Cell Wall:** Plant cells have a cell wall mainly made of cellulose.
- **Mode of Nutrition:** Most plants are autotrophic, synthesizing their own food through photosynthesis. However, a few members are partially heterotrophic, such as insectivorous plants (e.g., Bladderwort and Venus fly trap) or parasites (e.g., *Cuscuta*).
- **Body Organization:** They exhibit a tissue/organ level of body organization.
- **Included Groups:** Kingdom Plantae encompasses a diverse range of organisms, including algae, bryophytes, pteridophytes, gymnosperms, and angiosperms.
- **Life Cycle:** Plants exhibit two distinct phases in their life cycle: the diploid sporophytic phase and the haploid gametophytic phase. These phases alternate with each other, a phenomenon known as alternation of generations. The lengths and dependency of these haploid and diploid phases vary among different plant groups.
- **Further Study:** The characteristics of Kingdoms Plantae and Animalia are commonly referred to as plant and animal kingdoms and are dealt with separately in later chapters.

Viruses, Viroids, Prions, and Lichens

In Whittaker's five-kingdom classification system, there is no mention of lichens or acellular organisms such as viruses, viroids, and prions. These entities are briefly introduced because they do not fit neatly into the five kingdoms.

Viruses

- **Classification Status:** Viruses are not considered truly 'living' because they lack a cellular structure. They are non-cellular organisms.

- **Structure:** Outside a living cell, viruses exist as an inert crystalline structure. Once they infect a host cell, they take over its machinery to replicate themselves, often killing the host.

- **Historical Discovery:**

- **Dmitri Ivanowsky (1892)** recognized certain microbes as the causal agents of tobacco mosaic disease. He found them to be smaller than bacteria because they could pass through bacteria-proof filters.

- **M.W. Beijerinck (1898)** demonstrated that an extract from infected tobacco plants could cause infection in healthy plants. He named the pathogen "virus" and referred to the infectious fluid as "Contagium vivum fluidum" (infectious living fluid).

- **W.M. Stanley (1935)** showed that viruses could be crystallized, and these crystals primarily consist of proteins.

- **Composition:** Viruses are obligate parasites. In addition to proteins, they contain genetic material, which can be either RNA or DNA, but never both. The genetic material is infectious.

- Viruses infecting plants generally have single-stranded RNA.

- Viruses infecting animals can have single or double-stranded RNA or double-stranded DNA.

- Bacteriophages (viruses that infect bacteria) are typically double-stranded DNA viruses.

- **Protein Coat:** The genetic material is protected by a protein coat called a capsid. The capsid is made of small subunits called capsomeres, which are arranged in helical or polyhedral geometric forms.

- **Diseases:** Viruses cause various diseases in humans, animals, and plants. Examples include mumps, smallpox, herpes, influenza, and AIDS in humans. In plants, symptoms can include mosaic formation, leaf rolling and curling, yellowing and

vein clearing, dwarfing, and stunted growth.

Viroids:-

- **Discovery:** In 1971, T.O. Diener discovered a new infectious agent that was smaller than viruses and caused potato spindle tuber disease.

- **Composition:** Viroids were found to be free RNA. They lack the protein coat that viruses possess, hence the name "viroid". The RNA of the viroid has a low molecular weight.

Prions

- **Nature:** In modern medicine, certain infectious neurological diseases were found to be transmitted by an agent consisting of abnormally folded protein.

- **Size:** These agents are similar in size to viruses.

- **Diseases:** Notable diseases caused by prions include bovine spongiform encephalopathy (BSE), commonly known as mad cow disease in cattle, and its analogous variant, Creutzfeldt-Jacob disease (CJD) in humans.

Lichens

- **Nature:** Lichens are symbiotic associations, meaning they are mutually useful associations, between algae and fungi.

- **Components:**

- The algal component is called the phycobiont, and it is autotrophic (prepares food for the fungi).

- The fungal component is called the mycobiont, and it is heterotrophic (provides shelter and absorbs mineral nutrients and water for the algae).

- **Association:** Their association is so close that one might not realize they consist of two different organisms when observed in nature.

- **Pollution Indicators:** Lichens are considered very good pollution indicators because they do not grow in polluted areas

